

TONGDAO, China

WMO: 578450

Lat: 26.1588N

Lon: 109.7777E

Elev: 1302

StdP: 14.02

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.5	31.5	19.4	15.3	40.7	23.3	18.5	38.7	14.9	61.0	12.7	52.9	3.4	0	0.334

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	92.6	75.9	90.8	75.5	88.9	75.1	78.8	87.0	77.9	85.7	77.2	84.5	6.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.9	147.0	81.6	76.0	142.7	81.0	75.3	139.2	80.5	43.5	87.1	42.7	85.9	41.9	84.7	90.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.5	12.2	10.6	26.2	96.7	2.1	1.6	24.7	97.8	23.5	98.7	22.3	99.6	20.8	100.8		
			25.3	80.2	2.0	2.6	23.9	82.1	22.7	83.6	21.6	85.1	20.1	87.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.7	43.0	47.3	54.5	64.6	71.6	77.3	80.8	80.2	74.8	65.9	56.5	46.7
	DBStd	14.49	7.01	8.87	8.01	7.40	5.27	4.12	3.03	3.56	5.48	6.00	6.78	5.78
	HDD50	579	238	147	43	2	0	0	0	0	0	0	18	131
	HDD65	2521	681	499	339	97	8	0	0	0	4	62	265	566
	CDD50	5579	22	71	182	440	671	818	955	937	745	493	214	31
	CDD65	2045	0	3	13	84	214	368	490	472	299	90	11	1
	CDH74	16273	0	14	65	504	1395	2781	4607	4166	2195	491	54	1
	CDH80	5521	0	1	6	90	377	794	1683	1651	814	103	2	0
	WSAvg	4.3	4.1	4.6	4.6	4.7	4.3	4.4	5.0	4.0	3.8	3.7	3.8	3.8
	PrecAvg	55.8	2.2	2.6	3.8	6.7	9.3	9.1	5.6	5.3	3.0	3.5	3.0	1.7
Precipitation	PrecMax	71.1	4.4	5.5	7.6	11.4	16.0	19.5	18.5	18.7	8.3	7.8	10.2	4.0
	PrecMin	39.6	0.3	0.5	1.6	2.6	3.0	2.4	1.1	0.8	0.1	0.5	0.2	0.1
	PrecStd	8.7	1.0	1.2	1.4	2.4	3.7	3.9	4.1	4.0	2.0	1.9	2.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.3	76.5	80.0	85.7	89.7	91.9	94.7	95.0	92.9	87.2	79.1	70.2
		MCWB	58.1	63.4	67.5	70.1	71.9	75.0	77.1	76.7	74.4	70.4	66.7	59.8
	2%	DB	63.6	71.4	75.2	82.6	86.8	89.2	92.3	92.7	90.6	82.8	75.4	65.4
		MCWB	55.1	61.2	64.8	69.3	71.0	74.7	76.6	76.3	74.1	68.7	63.5	54.4
	5%	DB	58.7	67.2	71.5	79.6	84.3	87.0	90.3	90.8	88.1	79.6	72.0	60.8
		MCWB	51.9	59.2	62.6	68.7	71.6	74.8	76.4	75.9	73.7	67.1	61.8	52.2
	10%	DB	53.7	61.8	67.3	76.6	81.5	84.7	88.1	88.5	85.1	76.1	68.1	56.0
		MCWB	48.8	55.5	60.7	68.0	71.4	75.0	76.1	75.8	72.7	66.1	60.0	49.0
	0.4%	WB	60.9	64.7	69.1	73.8	77.2	78.8	80.0	80.0	77.9	74.4	69.6	63.7
		MCDB	66.0	74.3	77.4	81.0	82.6	84.9	88.5	88.5	86.6	81.5	76.3	67.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	56.9	62.5	66.9	71.9	75.6	77.6	78.8	78.8	76.6	72.3	66.8	57.1
		MCDB	62.0	69.6	73.7	79.2	81.8	84.0	87.2	87.7	85.0	78.5	72.6	62.8
	5%	WB	52.6	60.1	64.4	70.7	74.4	76.9	78.0	77.9	75.6	70.2	63.9	53.0
		MCDB	57.2	66.3	69.9	77.3	80.6	83.3	85.8	86.1	83.4	75.7	69.2	58.2
	10%	WB	49.1	56.2	61.8	69.2	73.0	76.1	77.2	77.1	74.4	68.3	61.1	50.1
		MCDB	52.8	61.2	65.8	75.2	78.8	82.1	84.6	84.6	81.8	73.8	65.5	54.8
Mean Daily Temperature Range	5% DB	MDBR	9.9	10.8	11.7	13.0	13.6	11.7	12.9	14.6	15.7	14.0	14.3	12.9
		MCDBR	22.5	20.3	21.2	20.0	21.0	16.9	18.1	19.0	20.9	21.5	22.5	25.0
		MCWBR	13.1	11.0	10.3	7.9	7.4	5.1	4.9	5.0	6.3	8.2	10.7	14.0
	5% WB	MCDBR	16.9	17.2	15.4	14.2	13.3	11.7	14.7	15.9	16.4	15.0	15.6	19.0
		MCWBR	11.1	10.3	9.3	7.2	6.1	4.6	4.8	5.2	5.6	7.4	8.8	11.8
Clear-Sky Solar Irradiance	taub		0.622	0.643	0.737	0.706	0.599	0.576	0.540	0.601	0.633	0.686	0.603	0.608
	taud		1.604	1.611	1.483	1.588	1.857	1.939	2.041	1.853	1.746	1.581	1.704	1.649
	Ebn at Noon		194	205	196	208	232	236	244	229	216	192	199	191
	Edh at Noon		74	78	94	86	66	60	55	65	71	80	66	68
All-Sky Solar Radiation	RadAvg		495	609	769	977	1114	1139	1508	1420	1210	947	764	606
	RadStd		128	155	115	143	182	160	146	111	133	117	150	143

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.47	N/A	N/A	+0.70	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+0.59	N/A	N/A	+1.13	N/A	N/A	N/A	N/A	+212	+128

Nomenclature: See separate page